Examiner only

2. Inos is a Welsh caravan manufacturer based in Denbighshire.
The materials used to build modern caravans are very different to those used in early caravans.

Table 1 shows how the materials used in manufacturing the body of a caravan have changed over time.

Year	Materials used
1920	wood
1950	steel and wood
1970	aluminium and wood
2015	glass reinforced plastic (GRP), polyester

Table 2 shows information about some of the materials used to make the body of caravans.

Material		Density (g/cm^3)	Stiffness (GPa)	Melting Point (°C)	Strength (MPa)	Is it brittle?	Does it corrode?
wood		0.6			1 000	No	no but rots when wet
steel		7.8	210	1 357	1 200	No	yes producing rust
aluminium		2.7	69	660	90	No	yes producing a resistant coating
GRP	glass fibres	2.4	180	1 400	3 500	Yes	No
	epoxy resin	1.5			60	Yes	No
polyester		1.9	150	121	250	Yes	No

Use the information above to answer the following questions.

(a) Describe how the weight and strength of caravan bodies changed between 1920 and 1970. [4]

Weight: ....

....

....

Strength: ....

....

....

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(b) (i) State **two** advantages of the materials used in 2015 over those used previously. [2]

1. ....

2. ....

(ii) State **one** disadvantage of the materials used in 2015 over those used previously. [1]

....

(c) GRP is a common composite material, in which glass fibres are mixed with epoxy resin.

(i) A GRP panel contains a volume of 300 cm^3 of epoxy resin.

Use the equation: [3]

$$\text{mass} = \text{density} \times \text{volume}$$

to calculate the mass of the resin.

mass = ... g

(ii) I. State the strength of glass fibre in N/cm^2. [2]

$$(1\,\text{MPa} = 100\ \text{N}/\text{cm}^2)$$

strength = ... N/cm^2

II. The cross-sectional area (csa) of a glass fibre is 0.00015 cm^2.

Use your answer in part (c)(ii)I. and the equation:

$$\text{force} = \text{strength} \times \text{csa}$$

to calculate the force required to break it. [2]

force = ... N

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